EXPERIMENT 19:

Perform Edge detection using Sobel Matrix along XY axis

AIM:- Edge detection using Sobel Matrix along xy axis

PROGRAM:-

```
import cv2

import numpy as np

import os

image_path = r'C:\Users\pavan\OneDrive\Opencv\house.png'

if not os.path.exists(image_path):

    print("Image file not found at:", image_path)

    exit()

img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

if img is None:

    print("Failed to load image.")

    exit()

sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)

sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

magnitude = cv2.magnitude(sobelx, sobely)

magnitude = cv2.convertScaleAbs(magnitude)

cv2.imwrite('SobelXY.jpg', magnitude)

cv2.imshow('Sobel XY-axis Edges', magnitude)

cv2.waitKey(0)

cv2.destroyAllWindows()

print("Sobel XY-axis edge detection complete. Output saved as SobelXY.jpg")
```

INPUT:-

Output:-

RESULT:-

Thus, the program is executed successfully